

CAST Imaging — Requirements Builder

Answer the questions on the left to generate the exact architecture & installation prerequisites for **your** deployment — not a generic checklist.

Your CAST Imaging Deployment Requirements

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Generated live from the answers on the left. Reference structure: doc.castsoftware.com/imaging/install .

scale: **Small Team**

users: **10–50 users**

platform: **Docker**

scenario: **Full (analysis + Viewer + Dashboards)**

topology: **Single machine**

egress: **Direct outbound**

auth: **Local accounts**

Validation notice. This tool derives recommendations from CAST Imaging's published documentation structure and standard CAST deployment practices. Always confirm exact minimum versions, default ports, and sizing against the current doc.castsoftware.com/imaging/install for the CAST Imaging release you are deploying.

1. Hardware sizing for your profile

Component	vCPU	RAM	Disk
All-in-one host (imaging-services + analysis-node + imaging-viewer + dashboards + PostgreSQL + Neo4j)	8	32 GB	300 GB

Neo4j (bundled with imaging-viewer) needs **16 GB RAM minimum, 32 GB recommended** — it defaults to an initial/maximum JVM heap of 12 GB, tunable via `imaging-viewer/.env` . Sizing above already accounts for this.

CAST's absolute floor is **256 GB free disk per node**, on SSD or equivalent high-IOPS storage (SAN configured with SSD) — the figures above already sit at or above that floor for your profile, but treat 256 GB as a hard minimum on any node regardless.

CAST's RAM floor differs by topology: **32 GB minimum for a standalone/all-in-one machine** (everything on one host), versus **16 GB minimum per node in a distributed/multi-machine deployment** — 32 GB recommended per node there too. The figures above already respect these floors.

Operating system / runtime versions

Item	Requirement
Host OS	Linux, 64-bit (x64) only, glibc ≥ 2.28 — Server Core / non-glibc distros not supported
Supported distributions	RHEL-family: RHEL 8, RHEL 9, Rocky Linux 8, Rocky Linux 9 · Debian-family: Debian 11, Debian 12, Ubuntu 22.04 LTS, Ubuntu 24.04 LTS
Docker Engine / Compose	Docker Compose v2.x+ (the version embedded in current Docker Engine) — the installer runs "docker compose" (no hyphen); the legacy standalone "docker-compose" v1.x is not compatible
Installing user	Must be in the sudoers list — the Docker installer requires sudo privileges
/tmp requirement	The installer uses <code>/tmp</code> during installation and needs it mounted with exec permission — if your hardening baseline mounts <code>/tmp</code> noexec (common on RHEL/CIS-hardened images), remount it exec or point the installer at an alternate temp dir with exec allowed
SELinux	Treat enforcing mode on RHEL/Rocky as a risk for bind-mounted volumes (missing <code>:z</code> / <code>:Z</code> context) — verify against requirements/software

Storage locations

CAST Imaging uses **five main storage folders** per its documented model (deploy, delivery, common-data, plus database/graph data). On Docker/Podman, the confirmed defaults are:

- `/opt/cast/shared/delivery` — reference information about applications and their source-code deliveries
- `/opt/cast/shared/common-data/<custom-folder>` — recommended location if you point analysis at a source-code folder directly; the analysis-node container already has access here

Full paths for every folder, plus Windows- and cloud-specific storage layouts, are at CAST's reference pages: [storage-locations](#), [.../docker](#) — not independently verified here (network egress to doc.castsoftware.com is blocked).

Client-side requirements

Profile	Requirement
End-user workstation (10–50 users)	Chrome / Edge / Firefox on Windows, macOS, or Linux — CAST regularly tests against current browser versions rather than pinning a strict minimum, so keep browsers up to date; 8 GB RAM, 1920×1080 recommended for architecture/dependency graphs
Delivery / analyst workstation	Same as above, plus local disk for source checkouts, a compatible JDK (≥ 17 LTS and ≤ 21, 64-bit, with <code>JAVA_HOME</code> set), and SCM client tools for the repositories selected in Section 8

2. Database requirements

PostgreSQL (RDBMS)

Item	Your configuration
Engine	PostgreSQL — port 5432 (the only RDBMS engine supported by CAST Imaging; stores analysis and persistence data)
Version	The bundled Docker image ships PostgreSQL 15 — pinned by CAST, nothing for you to choose. The legacy CAST Storage Service is end-of-life — deployments still on it must migrate to standard PostgreSQL. Confirm the full supported/minimum version range against doc.castsoftware.com/imaging/install/requirements/db before provisioning.
Hosting	Co-located
Character set	UTF-8 required; case-sensitive collation as specified by CAST for your CAST Imaging release
High availability	Optional — plan regular automated backups regardless

Neo4j (graph store)

Item	Your configuration
Requirement	Bundled with imaging-viewer — not a user-selectable engine; an ETL loads analysis results from PostgreSQL into Neo4j for the Viewer
Ports	7687 (Bolt) / 6362 (backup)
Hosting	Co-located with imaging-viewer.
Memory	16 GB RAM minimum, 32 GB recommended — defaults to a 12 GB initial/maximum JVM heap, tunable via <code>imaging-viewer/.env</code>

3. Network ports & FQDN allowlist — tailored to your answers

Only the flows relevant to the platform, integrations, and access methods you selected are listed — no wildcards, no unused rows.

Source	Destination	Port	Protocol	Reason	
End-user browser	CAST Imaging Web	443	HTTPS	Console / Viewer UI access — HTTPS is recommended but not mandatory — CAST Imaging can also be served over plain HTTP	Recommended
End-user browser	CAST Imaging Web	8090	HTTP	Console / Viewer UI access without HTTPS, or redirect to HTTPS if enabled — CAST Imaging Web listens on 8090, not the standard port 80 — Use if HTTPS is not configured; otherwise redirect to HTTPS and disable	Conditional
End-user browser	Keycloak	8096	HTTP/HTTPS	CAST Imaging's embedded authentication/SSO broker — the browser is redirected here to sign in, whatever identity source (local, SAML, LDAP) is configured — If fronted by a reverse proxy, the proxy must forward this port/path too, not just CAST Imaging Web — otherwise sign-in breaks	Mandatory
End-user browser	Eureka	8098	HTTP/HTTPS	Service discovery/registry endpoint the browser/frontend accesses directly — If fronted by a reverse proxy, the proxy must forward this port/path too	Mandatory
Administrator workstation	CAST Imaging host (Linux)	22	SSH	Server administration	Mandatory
imaging-services	PostgreSQL (co-located on the single host)	5432	TCP	Application access to the PostgreSQL database	Mandatory

Source	Destination	Port	Protocol	Reason	
imaging-services	extend.castsoftware.com	443	HTTPS	Product updates and licensing artifacts	Mandatory
analysis-node	extend.castsoftware.com	443	HTTPS	Extensions catalog — technology analyzers used during analysis	Mandatory
CAST Imaging node(s)	registry-1.docker.io, auth.docker.io, production.cloudflare.docker.com	443	HTTPS	Pull CAST's Docker Hub images at install/update time — Exact hostnames per Docker's own documented pull path — confirm CAST doesn't front them with its own registry for your release	Mandatory
analysis-node	Source repository (Git/SVN/DevOps over HTTPS)	443	HTTPS	Source code retrieval for analysis delivery	Mandatory

4. HTTPS / TLS

HTTPS is recommended, not mandatory. CAST Imaging can be served over plain HTTP; enable HTTPS whenever the deployment is reachable beyond a fully trusted, isolated network segment.

- Certificate source: **CA-issued**.
- TLS termination: **External reverse proxy**.
- If you enable HTTPS, the customer/you must provide the certificate and private key — CAST Imaging does not issue public TLS certificates.
- If HTTPS is enabled, disable plain HTTP (port 8090 — CAST Imaging Web's actual HTTP port, not the standard port 80) once it is confirmed working, or keep only an HTTPS→redirect rule.
- On Docker, CAST's documented approach is a dedicated Nginx vHost `.conf` file with `ssl_certificate` / `ssl_certificate_key` directives, `ssl_protocols TLSv1.2` or higher, and server cipher preference enabled — not a CAST-specific certificate variable.
- **The private key must not be encrypted** ("BEGIN ENCRYPTED PRIVATE KEY") — CAST's bundled Nginx does not support encrypted keys; strip the passphrase before use.
- Self-signed certificates must be installed/trusted on every workstation that accesses CAST Imaging, or users will see browser warnings.
- Behind any reverse proxy: forward `X-Forwarded-For` , `X-Forwarded-Proto` , `X-Forwarded-Host` , and set `KC_PROXY=edge` , `KC_PROXY_HEADERS=xforwarded` , `PROXY_ADDRESS_FORWARDING=true` in `cast_imaging_services/.env` (and `cast_imaging_viewer/.env` if the Viewer is deployed) — otherwise redirects and asset URLs will be generated wrong.
- Serving CAST Imaging under a URL path prefix (e.g. `https://host/imaging`) additionally requires setting `APP_CONTEXT` and editing `cast_imaging_services/docker-compose.override.yml` — see CAST's reverse-

proxy-with-context-URL reference.

5. Authentication prerequisites

- No external identity provider required.
- Plan initial admin password rotation and manual account lifecycle management.
- Recommended even alongside SSO: keep a break-glass local admin account.

6. CAST Extend access & licensing

- Allow outbound HTTPS (443) from the CAST Imaging node to extend.castsoftware.com.
- Generate an **API key from the CAST Extend website** and configure it in CAST Imaging / CAST Extend client to authenticate extension downloads.
- Obtain a valid **CAST Imaging license file** from your CAST account team; confirm it covers the modules you selected (MCP/AI: no; Highlight integration: no).
- Store the license file and the API key in a secrets store — not in plain configuration files under version control.

7. Pre-installation checklist for your profile

- ☐ Server(s) provisioned meeting the sizing in Section 1 (Docker, Full (analysis + Viewer + Dashboards), Single machine)

- ☐ `curl` installed on every machine that runs the installer — required for the installer's own service health checks (CAST requirements precheck)

- ☐ Installing user confirmed in the sudoers list on every machine

- ☐ `/tmp` confirmed mounted with `exec` permission on every machine (remount or redirect if your hardening baseline sets `noexec`)

- ☐ PostgreSQL provisioned as: Co-located

- ☐ TLS certificate obtained (CA-issued) and TLS termination configured (External reverse proxy)

- ☐ All firewall rules from Section 3 opened before install day

- ☐ Outbound access to extend.castsoftware.com confirmed

- ☐ Access to Docker Hub confirmed (registry-1.docker.io, auth.docker.io, production.cloudflare.docker.com) for pulling CAST's images

- ☐ CAST Extend API key generated and stored securely

- ☐ CAST Imaging license file obtained, covering the selected modules

- ☐ End-user and delivery workstation requirements communicated to stakeholders
